

Lecture 21: Equilibrium figures

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Outline and Motivation

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- Today, we will investigate these equilibrium equations in more detail.
- We will see how the requirement of **hydrostatic equilibrium** places strong constraints on the shape and internal structure of a planet.
- This will lead to Clairaut's theory, which explains why slowly rotating planets are **oblate spheroids**.

The Equilibrium Equations

The general equilibrium equation for a steadily rotating, self-gravitating body is a balance of forces:

Equilibrium Force Balance

$$\underbrace{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m}_{\text{Centrifugal Force}} - \underbrace{\frac{\partial \sigma_{ij}}{\partial x_j}}_{\text{Stress Divergence}} - \underbrace{\rho g_i}_{\text{Gravitational Force}} = 0$$

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- This equation is general and holds for any material. Over geological timescales, planets behave like fluids, so we cannot assume they are purely elastic.
- For a general stress tensor σ_{ij} , this is an **underdetermined** problem (3 equations for 6 unknown stress components).

Hydrostatic Equilibrium (1/3): The Approximation

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- The stresses associated with mantle convection are much smaller than the hydrostatic pressure, so this is an excellent approximation for the Earth's global shape.

Hydrostatic Equilibrium (2/3): The Governing Equation

Substituting $\sigma_{ij} = -p \delta_{ij}$ into the equilibrium equation gives:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0$$

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We can simplify this by writing the forces as gradients of potentials.

- Gravitational force: $\mathbf{g} = -\nabla\phi$
- Centrifugal force: $\rho \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}) = \rho \nabla\psi$

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Hydrostatic Equilibrium Equation

We introduce the **gravity potential** $\gamma = \phi + \psi$. The equation then takes the simple form:

$$\nabla p = -\rho \nabla \gamma$$

Hydrostatic Equilibrium (3/3): Consequences

The equation $\nabla p = -\rho \nabla \gamma$ has profound consequences.

- It implies that the vector ∇p is parallel to the vector $\nabla \gamma$.
Therefore, surfaces of constant pressure (**isobars**) must coincide with surfaces of constant gravity potential (**equipotentials**).

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- Since the planet's surface is a surface of zero pressure, the **surface of a planet in hydrostatic equilibrium must be an equipotential surface**.
- By taking the curl of the equation ($\nabla \times (\nabla p / \rho) = \mathbf{0}$), we find:

$$\nabla \rho \times \nabla \gamma = \mathbf{0}$$

This implies that surfaces of constant density must also coincide with the equipotential surfaces.

Conclusion

For a planet to be in hydrostatic equilibrium, its surfaces of constant pressure, density, and gravity potential must all coincide.

Slowly Rotating Planets (1/3): Perturbation Setup

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- For a non-rotating planet, the only hydrostatic solution is a spherically symmetric one.
- For the Earth, the centrifugal potential is very small compared to the gravitational potential ($\sim 1/300$). This suggests we can treat rotation as a small perturbation.
- We expand all quantities as a spherical zeroth-order term plus a small first-order perturbation (denoted by subscript 1).

Perturbation Expansions

$$\rho = \rho_0(r) + s\rho_1(\mathbf{x})$$

$$p = p_0(r) + sp_1(\mathbf{x})$$

$$\phi = \phi_0(r) + s\phi_1(\mathbf{x})$$

$$\psi = s\psi_1(\mathbf{x})$$

The shape is also perturbed: $r = b + sh_1(\theta, \varphi)$.

Slowly Rotating Planets (2/3): The Linearised Equation

Substituting the expansions into the hydrostatic equilibrium equation and keeping only the first-order terms gives the linearised equation:

First-Order Hydrostatic Equilibrium

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = 0$$

where $\gamma_1 = \phi_1 + \psi_1$ is the perturbation to the gravity potential and $g_0 = d\phi_0/dr$.

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Boundary Condition

The condition that pressure is zero on the perturbed surface, $r = b + sh_1$, gives a linearised boundary condition on the reference surface $r = b$:

$$p_1 - \rho_0 g_0 h_1 = 0 \quad @ \quad r = b$$

Slowly Rotating Planets (3/3): Sketch of the Solution

The linearised equation can be manipulated to relate all perturbations to the gravity potential perturbation, γ_1 .

- Taking $\hat{\mathbf{r}} \times (\text{Eqn})$ shows that $\nabla(p_1 + \rho_0\gamma_1)$ is radial. Assuming perturbations average to zero on spheres, this gives:

$$p_1 = -\rho_0\gamma_1$$

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- Taking $\nabla \times (\text{Eqn})$ similarly shows that:

$$\rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1$$

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- The boundary condition then gives the shape perturbation:

$$h_1 = -\frac{\gamma_1}{g_0} \quad @ \ r = b$$

Main Result

All first-order perturbations (p_1, ρ_1, h_1) are directly proportional to the

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Conclusion: Clairaut's Theory

This procedure, first developed by Clairaut, allows one to calculate the shape and internal structure of a slowly rotating planet in hydrostatic equilibrium. The resulting shape is an **oblate spheroid**, flattened at the poles. For the Earth, the equatorial radius is about 21 km larger than the polar radius.

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- A non-rotating hydrostatic planet must be spherically symmetric.
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- This requires that its surfaces of constant pressure, density, and gravity potential ($\gamma = \phi + \psi$) must all coincide.
- A non-rotating hydrostatic planet must be spherically symmetric.
- The shape of a **slowly rotating** planet can be found using perturbation theory. The key result is that all perturbed quantities (pressure, density, shape) are proportional to the gravity potential perturbation.
- This leads to **Clairaut's theory**, which predicts the planet's shape to be an **oblate spheroid**.